

OR

Game theory

Decision Making Situations

- Deterministic
- Probabilistic
- Uncertainty

Terminologies:

players: A & B

Strategy: (a course of action the player might take)

- Pure strategy (one strategy only)
- Mix strategy (multiple strategies followed)

If the outcome of the payoff matrix is positive then
Gain to player A
Loss to player B

If negative then,
Gain to player B
Loss to player A

→ Maximin Principle:

Maximize the minimum guaranteed gain of Player A.

→ Minimax Principle:

~~Max~~ Minimize the maximum losses.

Saddle point:

where Maximin = Minimax value

~~No saddle pt~~

Fair game : where $V=0$

- value of the game :
If game has a saddle point then the value of the cell at the saddle point is called value of the game

- Two person zero sum game :-
If gain of one player is equal to the loss of another player then that game is called as a zero sum game

★ Pure Strategies

① Find optimum strategies of the players in the following games.

		B			Row Min
		1	2	3	
A	1	25	20	35	20
	2	50	45	55	45
	3	58	40	42	40
Col max		58	45	55	
↓ Minimax					

Since Maximin = Minimax $\therefore$ this game has saddle pt.
value of the game = 45
 $V=45$

optimal probabilities of

$$A [p_1, p_2, p_3] = A [0, 1, 0]$$

$$B [q_1, q_2, q_3] = B [0, 1, 0]$$

★ No saddle points = Mixed strategy
(Maximin ≠ Minimax)

★ Mixed Strategies:

Consider the payoff matrix with respect to player A and solve optimally:

	(B)	
(A)	9	7
	5	11

Low Min

(7) ← Maximin
5

col Max (9) 11

↑

Minimax.

here Maximin ≠ Minimax (7 ≠ 9)
(No Saddle point)

	<u>odds</u>		Prob			
<table><tr><td>9</td><td>7</td></tr><tr><td>5</td><td>11</td></tr></table>	9	7	5	11	6	$P_1 = 6/2+6 = 6/8$
9	7					
5	11					
	2	$P_2 = 2/2+6 = 2/8$				
<u>odds</u>	4	4				

~~Prob~~

Prob $q_1 = \frac{4}{8}$ $q_2 = \frac{4}{8}$

Ans

A $\rightarrow$ maximize payoff
B $\rightarrow$ minimize payoff

* Principle of dominance :

When a matrix has no immediate saddle point or is larger than 2×2 , reduce its dimension by eliminating inferior strategies.

Rules :

1) Row Dominance (Player A) : If every entry in row i is greater than or equal to the corresponding entry in row k ($\text{Row } i \geq \text{Row } k$), Row k is dominated and can be deleted.

2) Column Dominance (Player B) : If every entry in col j is less than or equal to the corresponding entry in ~~row~~ col m ($\text{Col } j \leq \text{Col } m$); col m is dominated and can be deleted (since B wants to minimize loss).

3) Modified/Average dominance : A row/col can be eliminated if its dominated by the average (or weighted avg) of 2 or more other rows/cols.

Mixed (2x2)

	B_1	B_2	prob
A_1	5 (a_{11})	1 (a_{12})	p
A_2	3 (a_{21})	4 (a_{22})	$1-p$
	q	$1-q$	

p = probability Player A plays A_1 and q is the probability player B plays B_1 .

Formulae: $p = \frac{a_{22} - a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$

$$= \frac{4 - 3}{(5 + 4) - (1 + 3)}$$

$$= \frac{1}{9 - 4} = \frac{1}{5}$$

$$= \frac{1}{5} = \frac{1}{5}$$

$$\therefore 1 - p = \frac{4}{5}$$

$$q = \frac{a_{22} - a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$$

$$= \frac{4 - 1}{(5 + 4) - (1 + 3)}$$

$$= \frac{3}{9 - 4} = \frac{3}{5}$$

$$\therefore 1 - q = \frac{2}{5}$$

$$V = \frac{a_{11}a_{22} - a_{12}a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})} = \frac{20 - 3}{5} = \frac{17}{5}$$

~~3.4~~

3.4

Maximize

Questions :

(1) Solve the game:

		B				
		I	II	III	IV	V
A	1	-2	5	-3	6	7
	2	4	6	8	-1	6
	3	8	2	3	5	4
	4	15	14	18	12	20

$$\text{Row min}_1 = \min[-2, 5, -3, 6, 7] = -3$$

$$\text{Row min}_2 = \min[4, 6, 8, -1, 6] = -1$$

$$\text{Row min}_3 = \min[8, 2, 3, 5, 4] = 2$$

$$\text{Row min}_4 = \min[15, 14, 18, 12, 20] = 12$$

$$\text{Now Maximum}[-3, -1, 2, 12] = 12$$

$$\therefore \text{Maximum}(\underline{V}) = 12$$

$$\text{Col Max [I]} = \max[-2, 4, 8, 15] = 15$$

$$\text{Col max [II]} = \max[5, 6, 2, 14] = 14$$

$$\text{III} = \max[-3, 8, 3, 18] = 18$$

$$\text{IV} = \max[6, -1, 5, 12] = 12$$

$$\text{V} = \max[7, 6, 4, 20] = 20$$

$$\text{Now minimum}[15, 14, 18, 12, 20] = 12$$

$$\text{Maximum}(\underline{V}) = 12$$

$$\therefore \text{Maximum} = 12 = \text{Minimum}$$

$\therefore$ This game has a saddle point corresponding to entry.

row 4, col IV

$$\text{Value of game} = \underline{12}$$

$\therefore$ optimal strategy $A = 4$ $[0, 0, 0, 1, 0]$
 " " $B = 10$ $[0, 0, 0, 1, 0]$

$\therefore V$ is positive

$\therefore$ favorable for A

Game with no saddle point.

(a)

	B	
	3	5
A	4	2

→ Check for saddle point.

$$\text{Row min } A_1 = \min[3, 5] = 3$$

$$\text{Col max } B_1 = \max[3, 4] = 4$$

$$\text{Row min } A_2 = \min[4, 2] = 2$$

$$\text{Col max } B_2 = \max[5, 2] = 5$$

$$\text{Maximum}[3, 2] = 3$$

$$\text{Minimum}[4, 5] = 4$$

$$\therefore \text{Maximum}(V) = 3$$

$$\text{Minimax} = [V] = 4$$

$\therefore \text{Maximum} \neq \text{Minimax}$

$\therefore$ No saddle point.

$\therefore$ Mixed game.

	B	
S	3, 5	P
A	4, 2	1-P
	q, 1-q	

where P = probability that A_1 is played
 q = " " " B_1 is played.

Formulas:

$$P = \frac{a_{21} - a_{22}}{(a_{11} + a_{22}) - (a_{21} + a_{12})} = \frac{4 - 2}{(3 + 2) - (5 + 4)} = \frac{2}{4}$$

$$\therefore 1 - p = 2/4$$

$$q = \frac{3}{4}$$

$$1 - q = 1/4$$

$$V = \frac{(5 \times q) - (3 \times 7)}{4} = \frac{20 - 21}{4} = \frac{-1}{4} = -0.25$$

$\therefore$ ~~optimal~~ optimal strat A $\Rightarrow \begin{pmatrix} A_1 & A_2 \\ 0.5 & 0.5 \end{pmatrix}$

B $\Rightarrow \begin{pmatrix} B_1 & B_2 \\ 0.75 & 0.25 \end{pmatrix}$

$$\therefore V = -0.25$$

$\therefore$ favorable game for A.

(Q)
$$\begin{matrix} & \begin{matrix} B_1 & B_2 \end{matrix} \\ \begin{matrix} A_1 \\ A_2 \end{matrix} & \begin{pmatrix} 12 & 4 \\ 3 & 7 \end{pmatrix} \end{matrix} \quad \begin{matrix} p \\ 1-p \end{matrix}$$

$q \quad 1-q$

Saddle point :-

$$\text{Row min } A_1 = \min(12, 4) = 4$$

$$\text{" } A_2 = \min(3, 7) = 3$$

$$\text{Maximin}(V) = \max(4, 3) = 4$$

Maximin

~~(2) 6-4~~ ~~2-1~~ ~~2~~

Col max $B_1 = \max[12, 3] = 12$

Col max $B_2 = \max[4, 7] = 7$

Minimum

~~Maximum~~ ~~Maximum~~

Row Minimum $= 7$

$4 \neq 7$

$\therefore$ NO saddle.

$\therefore$ mixed

denominator $= (12+7) - (4+3) = 19-7 = 12$

$p = \frac{7-3}{12} = \frac{4}{12} = \frac{1}{3}$

$q = \frac{7-4}{12} = \frac{3}{12} = \frac{1}{4}$

$\therefore 1-p = \frac{2}{3}$

$1-q = \frac{3}{4}$

$V = \frac{84 - 12}{12} = \frac{72}{12} = 6$

$\therefore$ optimal strat $A \Rightarrow \begin{pmatrix} A_1 & A_2 \\ 0.33 & 0.667 \end{pmatrix}$

$B \Rightarrow \begin{pmatrix} B_1 & B_2 \\ 0.25 & 0.75 \end{pmatrix}$

$\therefore V > 0$

$\therefore$ favorable for A.

6) $\begin{pmatrix} 5 & 4 \\ 9 & 3 \end{pmatrix}$

$p = \frac{3+9}{12} = \frac{12}{12} = 1$

$8 - (4+9) = 8-13 = -5$

$8 - (-4+9) = 8-5 = 3$

$8 - (-13) = 21$

$V = 15 = \frac{36}{2.4}$

(Q)
$$\begin{matrix} & B_1 & B_2 \\ A_1 & \begin{bmatrix} 20 & 8 \end{bmatrix} \\ A_2 & \begin{bmatrix} -2 & 10 \end{bmatrix} \end{matrix}$$

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Check for saddle point.

Row min $A_1 = \min[20, 8] = 8$.

Row min $A_2 = \min[-2, 10] = -2$.

Now $\max(8, -2) = 8$

$\therefore \text{Maximin}(V) = 8$

Col max $B_1 = \max[20, -2] = 20$.

Col max $B_2 = \max[8, 10] = 10$.

Now $\min[20, 10] = 10$.

$\therefore \text{Minimax}(\bar{V}) = 10$

$\therefore \text{Maximin} \neq \text{Minimax}$
 $8 \neq 10$

$\therefore$ NO Saddle point

Mixed game.

	B_1	B_2	
A_1	20	8	P
A_2	-2	10	
	q	$1-q$	

Denominator = $(20+10) - (8 \times 2) = 30 - 16 = 14$

$P = \frac{10 - (-2)}{14} = \frac{12}{14} = \frac{6}{7}$

$1 - P = \frac{1}{7}$

$q = \frac{10 - 8}{14} = \frac{2}{14} = \frac{1}{7}$

$1 - q = \frac{6}{7}$

$$V = \frac{200 - (8x - 2)}{24} = \frac{200 + 16}{24} = \frac{216}{24} = 9$$

$\therefore$ Optimal strategies for A = $\begin{pmatrix} A_1 & A_2 \\ 0.5 & 0.5 \end{pmatrix}$
for B = $\begin{pmatrix} B_1 & B_2 \\ 1/12 & 1/12 \end{pmatrix}$

$\therefore V$ is +ve.
 $\therefore$ Favorable for A.

Principle of dominance sums

(Q1)

	B ₁	B ₂	B ₃	R ₄
A ₁	4	2	3	6
A ₂	3	4	7	5
A ₃	6	3	5	4

Check saddle point:

Rowmin A₁ = min[4, 2, 3, 6] = 2

Rowmin A₂ = min[3, 4, 7, 5] = 3

Rowmin A₃ = min[6, 3, 5, 4] = 3

Maximin = max[2, 3, 3] = 3

Colmax B₁ = max[4, 3, 6] = 6

= max[2, 4, 3] = 4

= max[3, 7, 5] = 7

= max[6, 5, 4] = 6

minimax (v) = 4

3 $\neq$ 4

NO Saddle point

Mixed

Now comparing B_1 & B_3 .

We can see that each element in B_3 is greater than each corresponding element in B_1 .
 $\therefore B_3$ gets dominated, delete B_3 .

New payoff matrix.

	I	II	IV
1	4	2	6
2	2	4	5
3	6	3	4

check saddle

$$\text{Max, min} = 3$$

$$\text{minimax} = 4$$

$$3 \neq 4$$

No saddle.

Now compare col II and IV,

Same reason as above

delete IV

New payoff

	I	II
1	4	2
2	2	4
3	6	3

Now saddle,

$$\text{Maximin} = 3$$

$$\text{Minimax} = 4$$

$$3 \neq 4$$

compare 1 and 3, 3 is less,
 $\therefore$ delete 1

I		II		
1	4	2		P
3	6	3		1-P

No saddle, $r+q$.

$$\text{denom} = (4+3) - (2+6) =$$

$$P = \frac{6-3}{-4}$$

I		II		
2	3	4		P
3	6	3		1-P

$$\text{denom} = (3+3) - 10 = 6 - 10 = -4$$

$$P = \frac{3-6}{-4} = \frac{-3}{-4} = 0.75$$

$$1-P = 0.25$$

$$q = \frac{3-4}{-4} = \frac{-1}{-4} = \frac{1}{4} = 0.25$$

$$1-q = 0.75$$

$$V = \frac{9 - 24}{-4} = \frac{+15}{-4} = \boxed{3.75}$$

Module ② Queuing Theory

Terms

- 1) customer
- 2) Queue
- 3) Service channel

Elements

- 1) Arrival pattern (pattern in which the customers arrive and join the system).

(random or fixed/static)

↓
[probability dist → Poisson dist]

Arrival rate: no. of customer arriving ^{unit-time} per unit time.
rate → per min or per hour.

Mean arrival rate = λ (frequency) (lambda)

eg: If 2 customer arrive in every 5 min

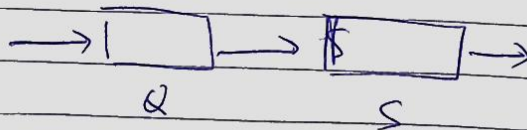
$$\therefore \lambda = \frac{2}{5} \text{ customers per min}$$

Mean time b/w arrival: $\frac{1}{\lambda}$

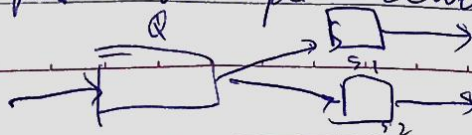
- 2) Service Mechanism or pattern :

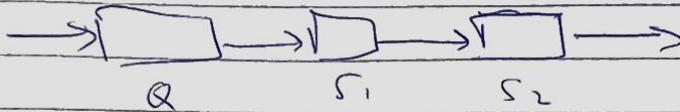
→ Arrangement server :

- ① Single queue single server :-

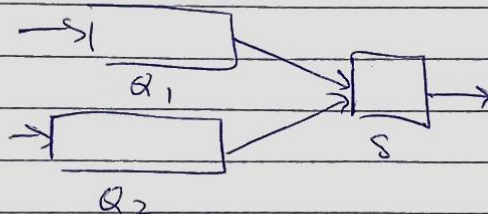


- ② Single queue multiple servers.

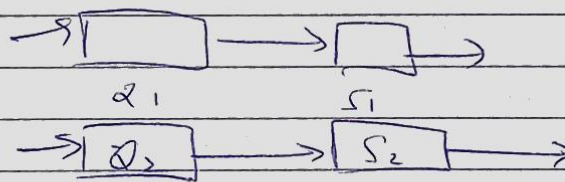




③ Multiple queue single server.



④ Multiple queue multiple server.



Service rate: no. of customers served per unit time.

mean service rate $\rightarrow \mu$

mean service time $\rightarrow 1/\mu$

③ Queue discipline: It refers to the order in which customer from queue are selected for service.

1 $\rightarrow$ FCFS or FIFO

2 $\rightarrow$ LCFS or LIFO

3 $\rightarrow$ SIRO (service in random order)

4 $\rightarrow$ Priority

④ Capacity of the system (Maximum no. of customers allocated in system)
 ↳ infinite
 ↳ finite.

⑤ Customer Behavior

- 1 → Balking - If queue is too long, customer doesn't join the queue.
- 2 → Reneging - Joins queue, but leaves due to long ~~time~~ ^{waiting time}.
- 3 → Jockeying - Switch b/w multiple ~~queues~~ ^{queues}.
- 4 → collusion - one customer joins on behalf of many.

⑥ Calling Population (∞ or finite)

Queue length (L_q)

System length (L_s)

Waiting time in queue (W_q)

Waiting time in system (W_s)

Utilization (or traffic intensity)

determines the usage of server

$$\rho = \frac{\lambda}{\mu}$$

(mean arrival rate / mean service rate)

Transient state

↳ If operating characteristics vary with time

Steady state

↳ If operating characteristics is independent of time.

★ Kendall's Notation

$(a/b/c) : (d/e)$

a: arrival distribution

b: service time distribution

c: no. of servers

d: max allowed no. of customers in system

e: queue discipline

Symbol for a & b

$M \rightarrow$ Markovian (Poisson/exponential)

$D \rightarrow$ Deterministic interval time

$E_k \rightarrow$ Erlang k-distribution.

if $k = 1 \rightarrow$ (exponential)

$k = \infty \rightarrow$ (deterministic)

GI = General independent arrival dist

GI = general departure distribution

Symbol for c $\rightarrow$ 1 = one server
s = multiple server

Symbol for d $\rightarrow \infty$
 $\rightarrow N$

Symbol for e
 $\rightarrow$ FCFS $\rightarrow$ LCFS
 $\rightarrow$ SIRO $\rightarrow$ mva

(mean arrival rate < mean service rate)

⇒ (M/M/1) (∞/FCFS)

~~$\mu - \lambda + \lambda^2$~~
 μ

Operating characteristics

1) P(server is busy)

$$P_b = \frac{\lambda}{\mu}$$

2) P(server is idle)

$$P_0 = 1 - \frac{\lambda}{\mu}$$

3) P_n (no. of customers in system) :

$$P_n = \left(\frac{\lambda}{\mu}\right)^n P_0 = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right)$$

4) P(no. of customers > given no. k)

$$P(n \geq k) = \left(\frac{\lambda}{\mu}\right)^k$$

$$\text{or } P(n > k) = \left(\frac{\lambda}{\mu}\right)^{k+1}$$

5) P(queue is non empty)

$$\begin{aligned} P(n > 0) &= 1 - P_0 = P_1 \\ &= 1 - \left(\frac{\lambda}{\mu}\right)^0 \left(1 - \frac{\lambda}{\mu}\right) = \left(\frac{\lambda}{\mu}\right)^1 \left(1 - \frac{\lambda}{\mu}\right) \\ &= 1 - \left(1 - \frac{\lambda}{\mu}\right) = \frac{\lambda}{\mu} \end{aligned}$$

$$= \cancel{\lambda} + \cancel{\frac{\lambda}{\mu}} + \cancel{\frac{\lambda}{\mu}} + \frac{\lambda^2}{\mu^2}$$

$$= \boxed{\frac{\lambda^2}{\mu^2}}$$

- 6) ~~avg or~~ Average (or expected) ~~trans~~ no. of customer in system.

$$\boxed{L_s = \frac{\lambda}{\mu - \lambda}}$$

- (x) avg no. of customer in queue

$$L_q = \boxed{L_s - \frac{\lambda}{\mu}} = \frac{\lambda}{\mu - \lambda} - \frac{\lambda}{\mu}$$

$$\boxed{L_q = \frac{\lambda}{\mu} \times \frac{\lambda}{\mu - \lambda}}$$

- (8) average waiting time in system

$$W_s = \frac{\text{Expected no. of customer in system}}{\text{arrival rate}}$$

$$= \frac{L_s}{\lambda} = \frac{1}{\lambda} \times \frac{\lambda}{\mu - \lambda} = \boxed{\frac{1}{\mu - \lambda}}$$

- (a) Average waiting time in queue

$$W_q = (W_s) - \text{time in service}$$

$$= \frac{1}{\mu - \lambda} - \frac{1}{\mu}$$

$$= \frac{1}{\mu - \lambda} \times \frac{\lambda}{\lambda}$$

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ii) Avg length of non empty queue:

$$\begin{aligned} L_n &= \frac{L_q}{P(n>1)} = \frac{1 \times 1}{n \cdot n-1} \\ &= \left(\frac{1}{n}\right)^2 \\ &= \frac{1^2 \times n^2}{n \cdot n-1} \\ &= \frac{n}{n-1} \end{aligned}$$

(Q) Arrival (of customer) $\Rightarrow \lambda = 10$ per hr
service rate $\Rightarrow \mu = 12$ per hr

service time = 4 min = $4/60 = 1/15$
 service rate = $\frac{1}{\mu} = 1/15 = \underline{\underline{15}}$

(i) avg customers in Queue.

$$LQ = ?$$

$$LS = \left(\frac{1}{n-1} \right) - \frac{1}{n}$$

$$\frac{1}{n-1} - \frac{1}{n} = \frac{10}{15-10} - \frac{10}{15}$$

$$= \frac{10}{5} - \frac{10}{15}$$

$$\begin{aligned} &= 2 - 2/3 \\ &= 4/3 \end{aligned}$$

$$= \boxed{1.33}$$

~~(n-1)/n~~

(2) waiting time $\rightarrow 10$ minutes. $\rightarrow \frac{10}{60} = \frac{1}{6}$
(By how much flow of arrival num.)
 $W_Q = \frac{1}{n-1} = \frac{1}{n}$

~~$$= \frac{1}{15-10} = \frac{1}{5}$$

$$= \frac{1}{5} = \frac{1}{5}$$

$$\rightarrow \frac{2}{15} = 0.133$$~~

~~$$10 = \frac{1}{n-1} - \frac{1}{n}$$

$$10 = \frac{1}{15-1} - \frac{1}{15}$$

$$10 + \frac{1}{15} = \frac{1}{15-1}$$

$$\frac{151}{15} = \frac{1}{15-1}$$

$$15-1 = \frac{15}{151}$$

$$15-15 = 1$$~~

$$\frac{1}{6} = \frac{1}{15-1} - \frac{1}{15}$$

$$\frac{1}{6} + \frac{1}{15} = \frac{1}{15-1}$$

$$\frac{7}{30} = \frac{1}{15-1}$$

~~10~~ $\frac{30}{7} = 15-1$

$$n = 15 - \frac{30}{7} = 10.7$$